Predicting Turtle Nesting Trends Using Artificial Neural Network(ANN)

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INTRODUCTION

- Artificial Neural Networks (ANNs) are mathematic models that mimic how our brain learns by recognizing patterns in data
- They make predictions using three steps: multiplication, summation, and activation.
- What makes ANNs powerful is their ability to learn complex relationships and adapt automatically without needing manual rules.
- They work well for both supervised and unsupervised learning.

OBJECTIVE

- To assess the performance of the ANN model in predicting sea turtle nesting trends.
- To compare predicted results with actual nesting data.
- To use evaluation metrics such as Mean Squared Error (MSE), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).
- To determine the model's accuracy, reliability, and potential areas for improvement.

PROBLEM STATEMENT

Turtle nest numbers change throughout the year due to environmental and seasonal factors, making prediction a challenge. This project builds an ANN model to improve accuracy and support smarter conservation planning.

METHODOLOGY

1 DATA PREPROCESSING

- Removed missing values
- Normalized features (standardization - z-score scaling)
- Encoded categorical variables (e.g., nested)

2 FEATURE SELECTION

- Target: Nest
- Input features: Species, Nested, Num eggs 2, Length, Width
- Removed irrelevant features

3 MODEL TRAINING & TESTING

Built an ANN model with:

- 1 input layer
- 3 hidden layer (with ReLU activation)
- 1 output layer
- Trained using backpropagation and Adam optimizers
- Monitored training and validation loss to avoid overfitting

4 PERFOMANCE EVALUATION

Metrics Used:

- MSE , RMSE , MAE , R^2 , MAPE

Visuals:

- Line Graph (Actual vs Predicted), Histogram (Model Evaluation), Residual Plot and Loss vs Epoch curve

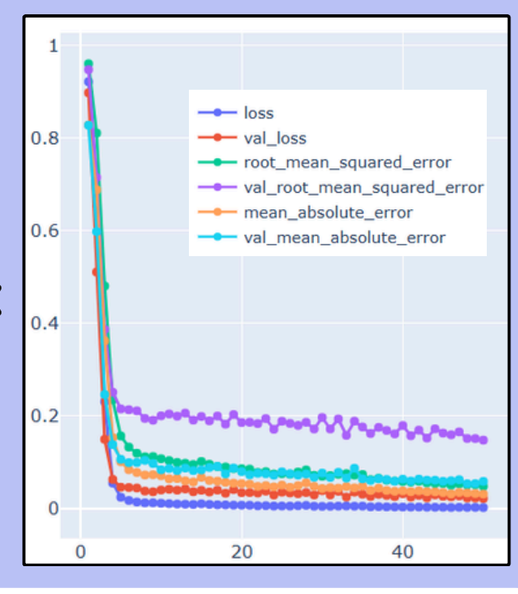
Interpretation:

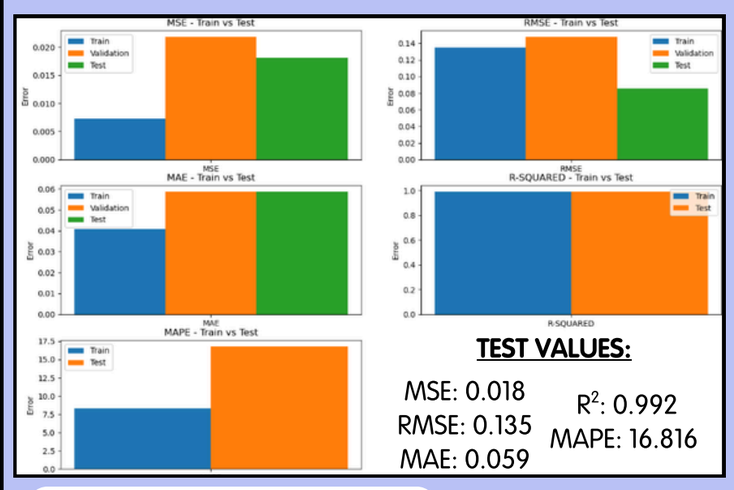
- The model accurately predicts nest counts, low error and strong generalization, which proves as a good performance

RESULT AND MODEL EVALUATION

Loss VS Epoch Curve

- All training metrics (blue, green, orange) drop sharply in the first few epochs and then stabilize, indicating the model is learning well.
- Validation metrics (red, purple, cyan) also decrease initially, showing the model is generalizing well.
- No sign of divergence between training and validation errors .
- No overfitting is observed.





TEST VALUES:

MSE: 0.018
RMSE: 0.135
MAE: 0.059
R^2: 0.992
MAPE: 16.816

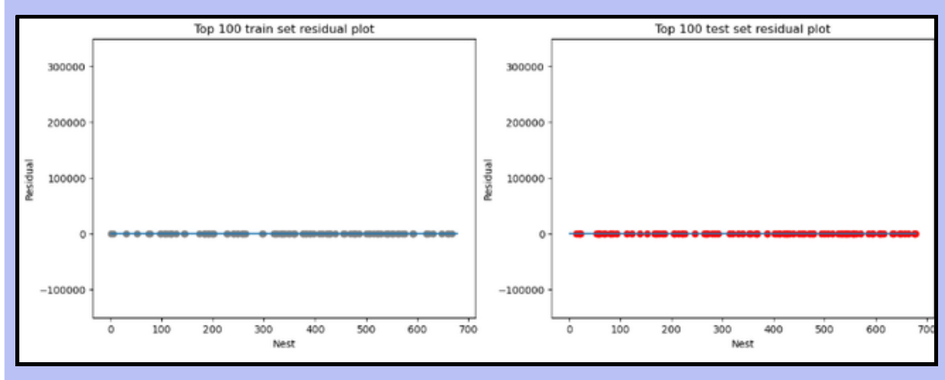
Model Evaluation Histogram

- MSE, RMSE, and MAE are low for train, validation, and test sets, showing good prediction accuracy.
- The close alignment between train and test error values suggests good generalization and minimal overfitting.
- R^2 is very close to 1, indicating the model captures nearly all variation in nest counts while also suggesting very strong predictive power.

Residual Plot

🟦 On the left:
The training data residuals mostly stay close to zero. This means the model learned the training data well and made predictions that are close to the real values.

🔴 On the right:
The testing data residuals also stay around zero. This shows that the model can predict new data accurately, not just the training data.

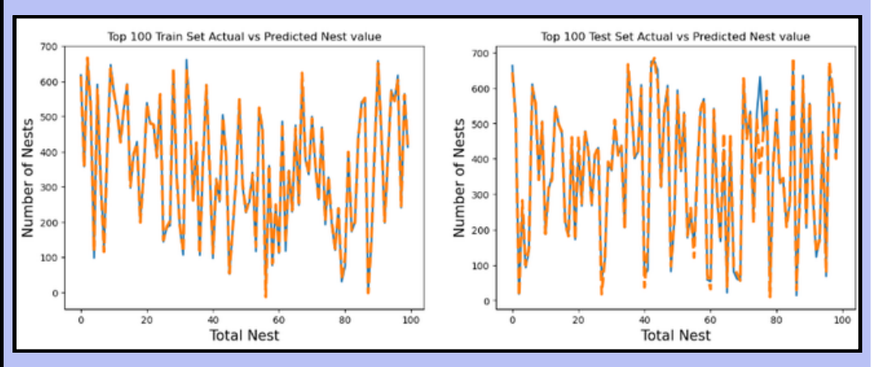


Model Performance:

Actual VS Predicted Nest Value

- Blue lines show actual nest counts; orange dashed lines show predicted values.

The predicted values (orange lines) closely follow the actual nest counts (blue lines). This means the model has learned the pattern well and can make accurate predictions, even on new data it has not seen before.



DATASET DESCRIPTION

- **Dataset Source**
  SEATRU, UNIVERSITI MALAYSIA TERENGGANU
- **Features**
  Raw Data Features : 12
  Features used in ANN : 7
  Number of Samples : 656
- **Target Variable**
  Nest

CONCLUSION

ANN was successfully used to predict the number of turtle nests based on various features. After cleaning and preparing the data, the ANN model was trained and evaluated.

- Loss plot showed the model learned well.
- Performance metrics (MSE, RMSE, MAE) were low.
- Residual plots showed small errors.
- Actual vs Predicted matched closely.

This shows that ANN can help monitor and predict turtle nesting patterns effectively.

REFERENCES

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